

Platform Operations Manual

1. Incident Response Overview

When a production incident is detected, the on-call engineer is paged automatically through the alerting system. The engineer has fifteen minutes to acknowledge the page before it escalates to the secondary on-call. Acknowledging an incident creates a dedicated channel where all communication about the issue must take place.

Severity levels range from SEV1, meaning full service outage affecting all customers, down to SEV4, meaning a minor cosmetic issue with no customer impact. SEV1 and SEV2 incidents require an incident commander to be assigned within ten minutes of detection.

2. Deployment Process

All changes to production services go through the standard deployment pipeline. Code is merged to the main branch after passing automated tests and receiving at least one approving review. Merging triggers a staging deployment automatically, followed by a manual promotion step to production once staging checks pass.

Deployments outside of business hours require explicit approval from a team lead unless they are fixing an active incident. Rollbacks can be triggered from the deployment dashboard and typically complete within two minutes for stateless services.

3. Monitoring and Alerting

Every service exposes a standard set of health metrics including request latency, error rate, and throughput. Dashboards are automatically generated from these metrics and linked from the service catalog. Alerts are configured against these dashboards using thresholds agreed upon during the service's initial design review.

Alert fatigue is taken seriously. Any alert that fires more than three times in a week without leading to action is flagged for review during the next reliability meeting, and either tuned, removed, or converted into an automated remediation.

4. On-Call Rotation

Engineers rotate through on-call duty on a weekly basis, with primary and secondary assignments published two weeks in advance. Swapping shifts is allowed with mutual agreement between engineers, recorded in the on-call scheduling tool so the paging system stays accurate.

New engineers do not join the on-call rotation until they have completed the incident response training and shadowed at least two live incidents with a senior engineer.

5. Disaster Recovery

Disaster recovery drills are conducted quarterly, simulating the loss of an entire region. During a drill, traffic is manually failed over to the backup region and the team measures the time to full recovery against the documented recovery time objective of thirty minutes.

Backup regions are kept in near real-time sync using asynchronous replication. Any drift beyond the defined replication lag threshold triggers an automatic alert to the infrastructure team.

6. Capacity Planning

Capacity reviews happen monthly, comparing current resource utilization against forecasted growth. Services trending above seventy percent sustained utilization are flagged for scaling review before they risk hitting hard limits during peak traffic.

Forecasts are built from a rolling twelve month trend combined with known upcoming product launches communicated by the product team during quarterly planning.

7. Security Review Process

Any new service or significant architectural change requires a security review before launch. Reviews cover authentication, data handling, and third party dependencies, and are conducted by a rotating member of the security team.

Findings from a security review are categorized by severity and tracked in the same issue tracker used for regular engineering work, with critical findings blocking launch until resolved.

8. Change Management

Significant infrastructure changes, such as database migrations or network reconfigurations, require a written change plan reviewed by at least two engineers outside the immediate team. The plan must include a rollback procedure and an estimate of customer impact.

Emergency changes made during an active incident are exempt from the standard review process but must be documented retroactively within twenty four hours.

9. Post-Incident Reviews

Every SEV1 and SEV2 incident receives a written post-incident review within five business days. The review focuses on timeline reconstruction and contributing factors rather than assigning individual blame, following a blameless postmortem approach.

Action items from post-incident reviews are tracked to completion and reported on monthly during the reliability review meeting attended by engineering leadership.

10. Vendor Management

Third party vendors providing infrastructure or data processing services are reviewed annually for security compliance and contractual SLA adherence. A vendor risk score is maintained and updated whenever a vendor reports an incident affecting their service.

New vendor relationships require sign off from both the security team and legal before any production data can be shared.

11. Data Retention

Customer data is retained according to the policy specified in the terms of service, typically ninety days after account closure unless a longer period is required by applicable law. Deletion requests are processed within thirty days and confirmed to the requesting customer by email.

Backups containing deleted customer data are purged on a rolling schedule to ensure they do not outlive the primary data retention window by more than one additional backup cycle.

12. Access Control

Production access is granted on a least privilege basis and reviewed quarterly by each team lead. Access to customer data specifically requires a documented business justification and automatically expires after ninety days unless renewed.

All production access changes are logged and available for audit through the internal access management dashboard, which is reviewed by the security team monthly.

13. Backup Procedures

Databases are backed up continuously using write ahead log shipping, with full snapshots taken daily and retained for thirty days. Restore procedures are tested monthly against a staging environment to confirm backup integrity.

Backup restore drills are timed and the results are compared against the documented recovery point objective to ensure the process still meets business requirements as data volume grows.

14. Service Level Objectives

Each customer facing service defines a service level objective for availability, typically ninety nine point nine percent measured over a rolling thirty day window. Error budgets derived from this objective are tracked on the reliability dashboard.

When a service exhausts its error budget for the month, new feature launches for that service are paused until the team completes a reliability focused sprint.

15. Escalation Procedures

If an incident is not resolved within the expected timeframe for its severity, it automatically escalates to the engineering manager and, for SEV1 incidents, to the director of engineering. Escalation does not remove the original on-call engineer from the incident.

Customers are notified proactively for any SEV1 incident lasting longer than thirty minutes, with updates posted to the public status page every fifteen minutes until resolution.

16. Communication Protocols

All incident communication happens in the dedicated incident channel to keep a single source of truth. Direct messages about an active incident are discouraged since they fragment the timeline needed for the post-incident review.

Status page updates use plain, non-technical language so customers without engineering backgrounds can understand the impact and expected resolution time.

17. Compliance Requirements

The platform maintains SOC 2 Type II certification, renewed annually through an independent audit. Engineers are required to complete compliance training during onboarding and annually thereafter.

Any change that affects data handling for regulated customers must be reviewed against the current compliance checklist before deployment to production.

18. Cost Optimization

Infrastructure costs are reviewed monthly by each team against a budget set during quarterly planning. Services exceeding budget by more than fifteen percent require a cost review before the following month's spend is approved.

Idle or oversized resources identified by the automated cost analysis tool are flagged to the owning team with a recommended right sizing action.

19. Documentation Standards

Every service must maintain a README covering ownership, architecture, and operational runbooks. Documentation is reviewed for accuracy whenever a significant change is made to the service, and stale documentation is flagged during quarterly service reviews.

Runbooks specifically must include step by step recovery procedures for the most common failure modes, written so that an engineer unfamiliar with the service could follow them during an incident.

20. Onboarding for New Engineers

New engineers complete a two week onboarding program covering the deployment pipeline, incident response process, and internal tooling. Onboarding concludes with a shadow on-call shift alongside an experienced engineer before joining the rotation independently.

Feedback from new engineers about the onboarding process is collected after thirty days and used to update the onboarding materials each quarter.